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Ingrid Kihlander, Mats Magnusson & Magnus Karlsson

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FEATURE ARTICLE

Critical Factors to Consider When Designing an Innovation Management System

The ISO 56002 standard offers a framework to design an innovation management system that can strengthen an organization's innovation efforts.

Ingrid Kihlander, Mats Magnusson and Magnus Karlsson

OVERVIEW: Organizations can implement an innovation management system in order to address the multidimensional challenges they often have in their practical innovation work. Until the ISO 56002 standard was released in 2019, there was no international standard for the design of innovation management systems. The standard provides support as a framework and highlights important systems elements. We studied two early adopters that have used this international standard to design their innovation management systems. Based on the study findings, we devised a practical approach for framing the design of an innovation management system. The approach includes a set of steps and critical considerations that include understanding a company's innovation ambitions and direction, analyzing the systemic dimensions of the system (the elements and how they interconnect), and introducing a balanced set of control mechanisms.

KEYWORDS: Innovation management system, Standard, ISO 56002, Innovation capabilities

An organization's ability to innovate—and the pace at which the innovation happens—is more imperative than ever. It is not a matter of if an organization should be concerned with its innovation capabilities, but rather how to deliberately induce, perform, and support innovation activities. Companies need to build long-term innovation capabilities and ensure they create and harvest value from their innovations.

Many organizations face common challenges in their innovation efforts. Often innovation happens sporadically rather than on a recurring or when needed basis. Furthermore, the innovation work is often limited in terms of which parts of the organization are involved and the methods and tools used (O'Connor et al. 2008). Another challenge is ensuring that the innovation activities from ideation to realized innovations are seamless to avoid the process slowing down or being blocked completely (Hansen and Birkinshaw 2007; Posner and Mangelsdorf 2017). Typically innovation initiatives are primarily started and driven by enthusiasts and entrepreneurial individuals. While highly driven and inherently motivated individuals going that extra mile are integral for many innovation activities, having such individuals is not an organizational feature and thus does not necessarily

Ingrid Kihlander is a senior researcher in innovation management at RISE Research Institutes of Sweden and an adjunct professor at the Swedish Defence University. She holds a PhD in machine design and an MSc in mechanical engineering from KTH Royal Institute of Technology in Stockholm. She has over 10 years of industry experience in change management and product development. Previously, she was director of the PIEp Product Engineering Innovation Program, a research and change program for increasing innovation capabilities, involving researchers, teachers, and students from five Swedish universities. She is also a research fellow at Institute for Management of Innovation and Technology (IMIT). ingrid.kihlander@ri.se

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Mats Magnusson is professor of product innovation engineering at KTH Royal Institute of Technology in Stockholm and permanent visiting professor at LUISS Guido Carli University in Rome. He is also vice head of the School of Industrial Engineering and Management at KTH, with responsibility for PhD studies. He has been a visiting professor at several universities abroad, is the chairman of the Continuous Innovation Network, and is active as a consultant and speaker in executive training programs on different innovation management matters. His research deals with strategic and organizational aspects of innovation, primarily in technology-intensive organizations. matsmag@kth.se

Magnus Karlsson is an adjunct professor in innovation management at KTH Royal Institute of Technology in Stockholm and project manager of innovation management systems at RISE Research Institutes of Sweden. He is a national expert and chair of the Swedish committee on innovation management at SIS Swedish Institute for Standards. He is a partner at Amplify and an advisor to multinational organizations in innovation management, a research fellow at the Institute for Management of Innovation and Technology (IMIT), and former director of new business development and innovation at Ericsson. He has a BSSC in international relations, an MSc in applied physics, and a PhD in technology and social change. magnus.p.karlsson@outlook.com

guarantee the existence of long-term innovation capabilities. Relying too much on individual efforts and skills may constitute a substantial risk especially if those key individuals decide to leave the organization.

Organizations may lack a clear direction or strategy for their innovation activities (Pisano 2015), which may lead to insufficient guidance for their innovation efforts. The result may be overly scattered and unstructured work, and organizations may not realize the full potential from their innovation activities. This multidimensional fragmentation limits the effects of the resources spent on innovation. It also points to the potential of establishing capabilities within an organization to work both systematically and systemically on innovation activities. Organizations need to develop more comprehensive, coherent, and consistent approaches for managing innovation activities. One solution is to deploy a systems approach for innovation management, including all necessary elements in the chain of innovation activities, spanning different functions and hierarchical levels, and covering both strategic and operational purposes.

A professionalization of the innovation management discipline is underway. Robbins and O'Connor (2023, p. 594) state that innovation management has "cleared the hurdles to becoming an emerging corporate profession," and highlight the international standard *ISO 56002:2019 Innovation management system – Guidance* (ISO 2019) as a tangible contribution to a codified body of knowledge in innovation management. A standard for innovation management systems may be useful as a supportive mechanism to address the aforementioned challenges—by providing a shared language for innovation and a framework that aims to support the management of innovation by taking a systems approach, encompassing strategic, structural, and cultural issues (Hyland and Karlsson 2021). Tidd (2021), however, has criticized the ISO 56002 standard for being too linear and general to provide support for different sectors, and that it lacks sufficient tools, which suggests that the standard by itself does not necessarily offer the intended support in designing innovation management systems.

Companies would benefit from empirical insights regarding whether and how ISO 56002 supports organizations in addressing the challenges of multidimensional fragmentation in innovation work. Due to its novelty, there is neither widespread awareness nor deployment of the ISO 56002 standard. Validated knowledge of actual usage of the standard and its effects is even more limited. In this study, we explore how organizations may use the ISO 56002 standard to design a suitable innovation management system to help accelerate innovation activities and build innovation capability. Our study addresses the following research question: What critical factors do organizations consider when developing an innovation management system based on the international guidance standard ISO 56002 for innovation management systems?

Literature Review

We consider innovation capability from a systems perspective and standards for innovation management systems.

A Systems Perspective on Innovation Capability

In 1986, Van de Ven published a seminal article on central problems in the management of innovation. He emphasized the importance of regarding innovation management with a systems perspective. Unfortunately, the systems perspective has not received the same attention as various innovation processes, methods, and tools, possibly because of its more abstract and comprehensive nature. Recent application of standards to the development of innovation management systems has prompted greater study.

Innovation capabilities highlight how an organization can use input in terms of ideas and knowledge and transform it into new value realized as products or other entities (Lawson and Samson 2001), where the combinations and configurations among the capabilities are crucial (Eisenhardt and Martin 2000). Lawson and Samson (2001) suggest adopting a systems approach for innovation capabilities that encompasses connected elements—for example, vision and strategy, directing relevant resources, customer and competitor intelligence, creativity and idea management, culture and climate, and organizational structures. O'Connor (2008) suggests a systems approach incorporating exploratory processes, relevant metrics, and mechanisms for collaboration and decision-making. However, a systems approach to innovation capabilities requires more than just identifying necessary system elements. For instance, Teece (2018) emphasizes that managers need to know much more about important relationships among the innovation management system elements. A particular challenge in designing a system for innovation is that innovation requires variation and mutation, whereas a quality management system aims to reduce variation (Reinertsen and Shaeffer 2005). Rather than optimizing a specific activity, innovation management systems have a dual purpose: they should support the efficient execution of innovation activities and simultaneously allow for challenging and continuously changing innovation's objectives. This dual purpose highlights the imperative of striking a balance between deploying supporting structures and fostering strategic independence and inducing diversity (Eisenhardt, Furr, and Bingham 2010).

Standards for Innovation Management Systems

In 2019, ISO published *ISO 56002:2019 Innovation management system – Guidance*, the first international standard for innovation management systems. Given that the standard was published less than five years ago, limited empirical research exists regarding its active deployment. Instead, current knowledge on innovation management standards is based mostly on national standards. In particular, the Spanish UNE 166002 R&D&I Management System Requirement (AENOR 2006) is the primary one investigated (Idris and Durmuşoğlu 2021). Research studies have shown that UNE 166002 positively influences innovative capabilities and business performance (Mir, Casadesús, and Petnji 2016), promotes all types of innovations (as in product, process, organizational, or marketing innovation) (Martinez-Costa, Jimenez-Jimenez,

and Castro-del-Rosario 2019), and also promotes both incremental and radical innovation (Giménez Espín, Martínez-Costa, and Jiménez Jiménez 2023). Although these studies point to innovation management systems as a fruitful means to create innovation, organizations have limited support for how to design an innovation management system using an international standard.

Method

The ISO 56002 standard serves as a principal contextual item in this study. Its content informed both data gathering and analysis. We used its seven key elements: *Context of the organization*, *Leadership*, *Planning*, *Support*, *Operations*, *Performance Evaluation*, and *Improvement* (Figure 1, Table 1).

We studied two early-adopter organizations that have used this ISO standard to design and implement suitable innovation management systems that in turn mitigate the multidimensional innovation challenges. Our selection criteria of explicit use of ISO 56002 limited the number of potential cases. Besides ISO 56002 being new, organizations using it are not registered—that is, they are not certified according to a requirement standard. We selected two organizations that suited our selection criteria. Company A is an engineering and development consultancy based in the United Kingdom that has received awards for its innovation excellence. Company B is an information and technology company based in Japan and has received awards as a global technology leader. Our selection covers a service-oriented company and a product-oriented company, based in different parts of the world. Both companies are large (over 15,000 employees) and operate in a global market.

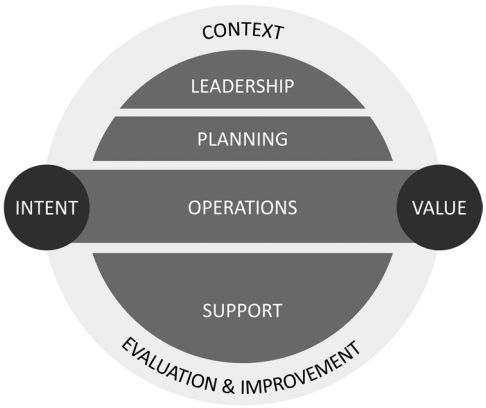


FIGURE 1. Key elements in ISO 56002, adapted from ISO (2019)

We used a case study methodology, including theoretical sampling, iterative organization of data, iterative comparison between data and theory, replicated single-case analyses, and a cross-case analysis (Eisenhardt 2021). This approach offered opportunities for data gathering in empirical settings regarding experiences using the ISO 56002 standard when designing an innovation management system.

We adopted a qualitative approach with semi-structured interviews with selected interviewees (Table 2), complemented by studying the companies' internal documentation. We conducted 14 online interviews from Company A, including staff in managerial positions, process facilitation, and the innovation management system core team. For Company B, we gathered data in two steps. Initially, we held a group interview with four innovation management officers, complemented by a round of written questions and answers to provide clarifications and add more detailed information. Then we conducted a second group interview onsite with three key staff from the first group interview. In the second interview, we asked participants about further development of the innovation management system, achievements, and perceived issues. We adopted slightly different approaches for data collection to be flexible and opportunistic (Eisenhardt 1989) in order to capture data from these early adopters. To ensure data consistency, the interview protocols were similar in terms of structure and terminology, with some minor adjustments to meet each specific setting. The interview questions

TABLE 1. Description of content in ISO 56002 Innovation management system: Guidance (adapted from Hyland and Karlsson [2021])

Key Elements in ISO 56002 Innovation Management System: Guidance	
Context	The organization should track external and internal issues and trends—for example, user preferences, technology developments, and internal capabilities—to identify opportunities and challenges that can trigger innovation activities.
Leadership	Based on the understanding of the context, senior management should demonstrate leadership and commitment by establishing an innovation vision, strategy, and policy, including the necessary roles and responsibilities.
Planning	Innovation objectives, organizational structures, and innovation portfolios should be established based on the direction set by senior management and the opportunities and risks identified.
Support	The support necessary for innovation activities should be put in place—for example, people with the right competences, financial and other resources, tools and methods, communication and awareness-creating activities, as well as approaches for intellectual property management.
Operations	Innovation initiatives or projects should be established in line with the company's innovation strategies and objectives. Innovation processes should be configured according to the types of innovations to be achieved and include the following generic innovation activities: identify opportunities, create and validate concepts, and develop and deploy solutions.
Evaluation	The innovation management system's performance should be evaluated regularly to identify strengths and gaps.
Improvement	Based on the evaluation, organizations should improve the innovation management system by addressing the most critical gaps with regard to understanding the context, leadership, planning, support, and operations.

TABLE 2. List of interviewees

Interviewees	Type of Interview
Company A	
Group Lead Innovation Management	Online (initial & follow-up)
Group Coordination Innovation Junior Staff	Online
Group Innovation Management Core Team, Support	Online
Consultant, Unit A	Online
Coordination Innovation Junior Staff, Unit A	Online
Innovation Lead, Unit B	Online
Consultant, Unit C	Online
Innovation Lead, Unit D	Online
Coordination Innovation Junior Staff, Unit D	Online
Innovation Lead, Unit E	Online
Coordination Innovation Junior Staff, Unit F	Online
Coordination Innovation Junior Staff, Group Support	Online
Global Sector Leader	Online
Company B	
Chief Innovation Officer (1)	Online, Group Interview 1
Innovation Management System Officer A	Online, Group Interview 1
Innovation Management System Officer B	Online, Group Interview 1
Innovation Management System Core Team	Online, group Interview 1
Senior Manager	Onsite, Group Interview 2
Chief Innovation Officer (2)	Onsite, Group Interview 2
Innovation Management System Manager	Onsite, Group Interview 2

probed for the intent to innovate, the system elements and how they relate to each other, critical considerations and choices made, as well as initial effects of the implemented innovation management system.

We analyzed the empirical data using processing and scrutiny techniques (Ryan and Bernard 2003). First, we used a priori themes primarily from ISO 56002 (Table 3) with a within-case focus to process the data into a tabular form (Eisenhardt 1989). Then we scrutinized the tabulated data iteratively to identify additional emerging themes—for example, in culture, risk appetite, openness for failure, and willingness to experiment emerged. Finally, we analyzed similarities and differences across the cases (Eisenhardt 1989) in order to draw conclusions on critical considerations in the design of an innovation management system and the practical applicability of the study findings.

Designing an Innovation Management System Using ISO 56002

In our study, we identified key considerations in designing an innovation management system related to both individual system elements and systemic considerations (that is, relationships among the elements). We present our findings using the key elements in ISO 56002. Each element contains several sub-elements. From the empirical analysis, we found that some sub-elements were more influential than others (noted in *italics*), and thus we focus on them here.

Context of the Organization

The *Context of the organization*—including external and internal factors influencing the innovation management system—were crucial considerations for both companies. It was important to determine the innovation intent and *Scope of the innovation management system* to lay a foundation for developing the innovation management system. This, in turn, was related to understanding the driving forces (the why) for implementing an innovation management system. Company A's motivation was a desire to enhance innovation capabilities across the organization and to level up from having "pockets of innovation." Company B identified in its strategy work that its current business would not be enough in the future and that it needed to increase its innovation capability.

Culture is a sub-element addressed from different angles in the data—for example, how risk-taking and experimentation need to be actively addressed through, for instance, promoting attitudes around managing uncertainties and learning from failures. Both companies indicated that leadership played an important role in building a non-blame culture. The interviewees from Company A indicated that the openness of the innovation processes has helped to build the culture. All of Company A's employees are welcome to enter proposals (opportunities, initiatives, or ideas) in the company's online portal. Since senior leaders then make the decisions to allocate resources to the proposals, there is a separation of ideas and people. Employees experience this separation as positive because it transfers risks from a personal level to the organization. For Company B, communication from senior management regarding risk-taking proved to be crucial.

The *Context of the organization*—including external and internal factors influencing the innovation management system—were crucial considerations for both companies.

TABLE 3. Key excerpts

Context	
Probing for understanding of the organizations' external and internal issues, needs and expectation of interested parties, collaborations, and culture	
Company A	**Company B**
"We have pockets of innovation excellence." (Innovation Unit Lead) "We need a system to help colleagues to progress all these fantastic values through to us, our clients, and society." (Group Lead Innovation)	"The movements in the industry as part of the current digital revolution threatens the foundations of the company's earnings. Changes are also underway in the company's relationship to our main customers, as they, too, experience changing business structures amid the social transformation." (Written response, after Group Interview 1) "To transform the corporate culture from a passive stance." (Group Interview 1)
Leadership	
Probing for leadership and commitment; focus on value realization; innovation vision, strategy, and policy; and roles and responsibilities	
Company A	**Company B**
"Having the group board members actually state that they support innovation, and they can see the value in having it—that is, a great enabler." (Group Innovation Core Team) "[Company A] has an overarching group strategy, cascaded into innovation plans, owned and delivered by the business units." (Group Lead Innovation)	"To ensure that the keen sense of resolution among top management is clearly conveyed, workshops are held by members of the top management, including the CEO. This is followed by training for managers at the division general manager level, then by training for managers at the department manager level." (Written response, after Group Interview 1)
Planning	
Probing for innovation objectives and planning to achieve them, organizational structures, and innovation portfolios	
Company A	**Company B**
"They [the board] wanted to ensure responsibility is with the operational business." (Group Lead Innovation) ". . . our business model [as consultancy] works on people being highly utilized on fee-earning projects . . . And that makes it really hard to look at how we can do things differently." (Innovation Lead Unit)	"A main characteristic of the [Company B] IMS is to have plans targeting challenging objectives." (Company documentation) "The division of work between the central innovation team and the business units posed a challenge regarding the need to achieve a comprehensive shared understanding of intentions regarding these opportunities between the innovation team and business units so that any promising concepts could be handed over smoothly to the business units." (Group Interview 1)
Support	
Probing for resources and time, competence, awareness and communication, and tools and methods	
Company A	**Company B**
"We have reserved budget and reserved time to look at this. The percent is not very high . . . But we have something available, and you have the ability to gain that resource." (Coordination Innovation Junior Staff Unit) "And then other clients have talked to us and said, in confidence, we are looking to apply for accreditation to the ISO 56002 standard because we want to demonstrate that we're serious about it. And it's grown our credibility." (Innovation Unit Lead)	"The manager is going to allocate the 10 percent of his time." (Group Interview 2) "These events have been held 100 times to date, with the participation of 2,972 individuals. This training is considered key for building a culture of innovation." (Written response, after Group Interview 1)
Operations	
Probing for innovation processes, identification of opportunities, creation and validation of concepts, development and deployment of solutions	
Company A	**Company B**
"Innovation [in Company A] has a process that is embedded into the business management system. If you want to know about innovation, how it works, that it is all laid out and everybody has access to that." (Group Innovation Core Team) "The framework includes process steps on discovery, incubation, upscaling, and consolidation as well as gates addressing opportunities, solutions, and reviews." (Company documentation)	"The process part of the system starts by identifying opportunities from the Sustainable Development Goals and analysis of related end-user needs." (Group Interview 1, confirmed in company documentation) "[Company B] has defined the IMS business process and IMS technology process based on the innovation process defined in the Chapter 8 of ISO 56002 [identify opportunities, create concepts, validate concepts, develop solutions, and deploy solutions]." (Written interview, company documentation)

(Continued)

TABLE 3. Continued.

Performance Evaluations, and Improvement	
Probing for monitoring, measuring, evaluation, deviations and corrective actions, and continual improvement	
Company A	**Company B**
". . . the [external assessment] demonstrates that we've got the processes in place not just to do the innovations, but to actually see some value out the other side. We've still got some work to do in terms of measuring that value. But what it does is it says to external parties—we don't just innovate for the sake of it, we innovate because we want to add value." (Innovation Unit Lead)	"Once every two months, the CINO [chief innovation officer] and the IPC [innovation promotion center] discuss with the individual responsible in each division to review and exchange information of both sides of innovation activities." (Written response, after Group Interview 1)
Systemic Considerations and Reflections	
Probing for relations among the elements, achievements of the innovation management system	
Company A	**Company B**
"Because if you haven't got your house in order, innovating is more difficult." (Global Sector Lead) "We've moved towards embedding innovation management processes, innovation culture. It's really encouraged some of those individuals to shine, and giving a few people some autonomy to go and develop these ideas has really brought out some fantastic individuals but also some fantastic leaders that have created teams around them." (Innovation Unit Lead)	"Training of employees we're aiming both for developing competence (for example, methods in design thinking) and to enhance the innovation culture." (Group Interview 1) "A major achievement is that we managed to go into the same direction in terms of managers as well as employees. So, we have changed. . . . We used to be quite different in terms of management, but gradually we changed that kind of situation. . . alignment among managers and employees—that is, working in the 'same direction.'" (Group Interview 2)

Leadership

The fact that *Leadership and Commitment* were considered important may not be surprising. For both companies, senior management's commitment to implement an innovation management system has been pivotal. Company B's senior management engaged beyond the supervisory level: they participated in the training and direct communication targeting employees. Interviewees from both companies emphasized their leaders' contributions to organizational culture, including fostering a willingness to embrace uncertainties and learn from failures.

Both companies have developed explicit *Innovation strategies* that provide direction to the innovation work, emphasize a *Focus on value realization*, and express long-term ambitions (even if the contents of the innovation strategies differ). Company A has chosen a more distributed way for its innovation work that gives the organizational units autonomy to operate their respective innovation work. Company A created a document hierarchy that outlines an overall group innovation strategy, which in turn informs innovation strategies and innovation plans for the different units within the organization.

Clarifications on *Organizational roles and responsibilities* related to innovation work have been key. Both companies appointed a core team to lead the development and implementation of their innovation management systems, but they have chosen different ways to assign responsibilities for the innovation work. Company A uses a more distributed model: the innovation management system core team at the group level is responsible for its innovation framework, including oversight of its online portal where employees post innovation proposals, some of which are ultimately developed. Company A's business units perform the operative innovation work, they adapt the framework to their respective settings (but use the company innovation terminology), and they assign people and budgets accordingly. This distributed unit-led model was an active choice after starting out with another distributed setup led instead by domain-expertise people (for example, energy or transport). The shift was a consequence of a perceived problem of ideas "getting stuck" in the first process step (for example, due to lack of budget). In Company A's distributed model, suggested responsibilities are assigned as part-time tasks combined with regular roles—for example, senior innovation leads assuming responsibility for budgeting and promoting innovation work, and junior staff supporting processes and ideas.

Company B assigns responsibility for innovation work differently. Its core team is responsible for company-level innovation capabilities, innovation management system implementation and improvements, and innovation training and awareness building (internally and externally). The core team also leads specific innovation initiatives in early phases—identifying opportunities and formulating and validating business model hypotheses—while the business units oversee development, business plan confirmation, and

For both companies, senior management's commitment to implement an innovation management system has been pivotal.

commercialization. In Company B, responsibility for the innovation work is divided along the innovation process.

Planning

Defining *Innovation objectives* and planning to achieve them is a crucial activity, but the companies differ substantially in their approach. In Company A's distributed model, individual business units set out plans and objectives. Being a consultancy firm, Company A's interviewees highlighted the challenges in carving out time for their internal innovation work, while always having their clients' value creation as a priority. Company B's explicit ambition to have the innovation management system spur breakthrough innovation is reflected in how it has set up challenging targets for its innovation work. Both companies work actively with *Innovation portfolios*: Company A distributes its innovation work, while Company B keeps it more centralized.

Support

The *Support* elements cover different aspects such as resources (for example, *Time*, *Finance*, and *Knowledge*). Allocating resources is critical for enabling innovation activities. Further, both organizations consider *Competence* critical but address it differently. At the time of the study Company A had just initiated intense work to formulate an innovation competency framework so it could clarify competencies needed and how to identify and develop people. From the outset Company B has considered training employees a core activity to feed its innovation management system. It launched an ambitious training program focused on skills for opportunity identification and the ability to design value propositions, and thousands of employees have taken part. Company B has an explicit target for managers to spend 10 percent of their time on innovation work.

Awareness and *Communication*, which are relevant for both management and employees, aim for understanding, ownership, and motivation; therefore, keeping the communication (and terminology) simple and clear is essential. External communication is highly relevant. Notably, Company B set out to communicate externally, aiming to enhance both partners' and employees' perception of the company as innovative. Both companies have used the trustworthiness they attach to the ISO brand in their own communications about their innovation management system.

Operations

Company A and B have designed their *Innovation processes* somewhat differently to suit their respective contexts and purposes. Both companies are using the ISO 56002 terminology and have given clarity in naming stages and activities to ensure that their employees understand the innovation processes, and to ensure both openness and availability of the process for the employees. Company A's employees are welcome to enter innovation proposals in the company's online portal in either (1) time-limited campaigns (starting with identifying opportunities) aimed to address challenges derived from market intelligence, or (2) in "always-on" (always open for proposals of all kinds). Company B strives for breakthrough innovations: its innovation process starts with identifying opportunities based on the United Nations Sustainable Development Goals and identified end-user needs, and then a technology process and a business process run in parallel.

Performance Evaluation and Improvement

Organizations use *Performance evaluation* to assess their current status. The companies' respective core teams for innovation management systems are responsible for *Improvement* and further development of the innovation management systems. Both companies emphasized the significance of such evaluations. Company A chose to be assessed against the ISO 56002 standard by a recognized, external certification actor. Preparations for the evaluation and the results proved to be valuable for Company A both internally and externally through both agreeing how to describe the innovation management system as well as confirming a certain level of innovation management proficiency. Company B performed a gap analysis during the initial design of the innovation management system. The results revealed a lower level of innovative culture than desired, which sparked a discussion on how to address this issue. This illustrates the value of having an evaluation result to communicate around and to base further actions on. Both companies stressed that the exact indicators to use for an evaluation are not always obvious, and that the indicators themselves will also need to be developed further.

Systemic Considerations and Reflections

The ISO 56002 standard has offered both companies support in addressing the many elements in their respective innovation management systems. This highlights a first critical systemic consideration—namely, to determine whether an organization has established all the relevant (and necessary) elements. Company A's innovation process is a centerpiece in all communications regarding its innovation management system. Company B has a visualization on how its innovation management system's elements are related—for example, *Understanding of context* and *Culture* serves as a foundation, and the *Innovation process* is supported by *Resources* and *Competence*.

Our data provide evidence of how the different elements interact and affect each other, which reveals a second systemic consideration—that is, to understand how the elements are related to each other and together create a system. For example, we can see how the openness of the *Innovation process* in Company A—whereby all employees can submit proposals and experience receiving support for ideas and for exploration—strengthens the *Culture*. In Company B, *Leadership* has the strongest effect on *Culture*—for example, the CEO discussed directly with employees the acceptance of failure and the importance of learning from these failures. While training employees in Company B primarily had the purpose of developing *Competence*—for instance, methods directly related to *Focus on value creation*—it also boosted the *Culture*. Another illustration of how elements affect each other is how the *Innovation strategy* provides direction but it

needs *Planning* and allocation of *Resources* to create actual value. Company A addressed this by shifting responsibility for the operative innovation work closer to the profit/loss centers in order to make the innovation activities happen.

A third systemic consideration is whether the innovation management system comprising interrelated elements has brought the companies in the right direction and thereby created alignment between the system effect and the overall ambition. When the companies reflect on the effects of their innovation management systems, they refer mainly to shifts in attitudes and behaviors. The empirical evidence shows that Company A's employees and managers discuss innovation more frequently, with more confidence, and more similarity across the company, which implies enhanced understanding and willingness to engage. The data show an alignment among Company B's managers and employees regarding innovation—that is, they are working in the "same direction." We also found a shift in Company B in terms of an enhanced openness for experimenting—in terms of employees' willingness and that management allows it—and a stronger understanding of customers.

Finally, both companies shared that external partners, clients, and employees now perceive them as more innovative companies.

Discussion

The use of a comprehensive innovation management system, aligned with the ISO 56002 standard structure, appears to address some of the central problems in innovation management highlighted by Van de Ven (1986) in a novel and potentially fruitful way. Innovation management theory has focused extensively on two fundamental problems in managing innovation. One issue concerns how organizations should develop ideas into new products and services, which arguably has been the focus of most innovation process research (Cooper and Friis Sommer 2020; Reinertsen and Shaeffer 2005; Van de Ven 1986). There also is the longstanding issue of how to deal with parts-whole relationships—as reflected in different organizational solutions that offer necessary levels of specialization and integration, primarily in R&D and marketing, respectively (Lawrence and Lorsch 1967; Van de Ven 1986). The innovation management systems in our study offer well-developed solutions to these two problems. What also emerged from the case study analysis is the fact that the innovation management system strengthens human attention to innovation and thereby increases the legitimacy of innovation activities and offers support for the necessary experimentation and risk-taking (McGrath 2010; Giaccone and Magnusson 2022).

The most significant differences in this instance compared to most previous innovation management approaches are institutional leadership's willingness to embrace learning from failures and the important role senior management plays in establishing firm-level innovation strategies that over time could reshape the firm's overall vision and purpose. To what extent these innovation strategies will be sustainable is not possible to foresee, as only recently have the two companies studied implemented their innovation management systems. Given the fact that firms may lack explicit innovation strategies, establishing tangible support for innovation is a positive development.

Our results suggest an innovation management system strengthens human attention to innovation, increasing the legitimacy of innovation activities and offering support for experimentation and risk-taking.

A final reflection concerns the risk of companies paying insufficient attention to continuous learning and to making needed changes in their innovation management system, as reflected in the ISO 56002 standard's *Improvement* element. If companies fail to adapt their innovation management systems to respond to changing innovation needs and conditions, their systems may ossify a set of outdated activities and structures, and possibly even hamper innovation.

A Practical Approach for Framing the Design of an Innovation Management System

The insights gained from studying early adopters from different sectors demonstrate that the ISO 56002 standard provides useful guidance for designing an innovation management system even if the standard does not offer any detailed company-specific guidance. Overall, we pinpointed three key considerations that are critical to helping organizations realize their innovation ambitions:

1. Understand innovation ambitions and develop a clear vision and strategy for innovation work.
2. Design the innovation management system to support the ambition and direction for the innovation work.
3. Establish management structures and control mechanisms that give the innovation management system enough energy, and balance contradictions and tensions sufficiently.

We outline important steps for each crucial consideration.

Innovation Ambitions, Vision, and Strategy

1. Define the intent of the innovation management system, and ensure a common understanding among both leaders and employees regarding the motivations for it.
2. Develop explicit and clear innovation visions and strategies. Consider whether your organization will benefit

from starting with one overall innovation vision and one overall innovation strategy.
3. Conduct an initial gap analysis or similar analysis to understand the current state and desired future states for all system elements. We recommend using the ISO 56002 structure, including systemic-level considerations about how the system elements interact and together build a value-creating system. This analysis can also serve as a tentative performance evaluation framework to develop further.

Innovation Management System Design

1. Identify and define the elements of your innovation management system, similar to how other management systems are normally presented. Using a comprehensive framework like the ISO 56002 standard that covers a comprehensive set of system elements and which is aligned with other management systems could simplify this task. Include a reflection or assessment of current and desired maturity levels of the elements. Are any elements missing? Remember to consider each element in a company-specific context. Consider how your organization's innovation management system may be presented internally to all employees in text and visually.
2. Identify how the elements interrelate and interact. Analyze the system to identify unwanted gaps and weaknesses such as counteracting elements, lock-ins, or deteriorating loops. Ensure that the system elements can reinforce each other.
3. Analyze the system elements to see how they align to support the organization's desired overarching innovation ambition. Is the system and its elements moving the company towards its specific innovation goals and objectives? Address what needs to be adjusted.

Management Structures and Control Mechanisms

1. Combine different types of control mechanisms. Identify control mechanisms relevant for the organization while at the same time ensuring sufficient autonomy to enable and allow for risk-taking and experimentation. Leaders should refrain from overly steering and controlling the process when starting to clarify and describe the innovation management system. On the other hand, efficiency and execution also have their place in an innovation management system, so it is important to ensure that mechanisms exist for introducing and realizing new values.
2. Ensure a long-term focus. This draws on several elements of the innovation management system, such as leadership, performance evaluation, and improvements. Have patience regarding achieving innovation results and in determining how to evaluate the innovation management system. Ensure that the initial version of the innovation management system your organization designs can be improved upon based on the insights gained along the way.
3. Appoint a core team responsible for "the map" of the innovation management system to enable the long-term focus and a systems view. Allowing the core team to be a focal point and to work with a long-term perspective enables the organization to aggregate different enhancements in the innovation work and thereby accelerate capabilities for innovation. Ensure senior management's support for the task.

For organizations interested in applying a systems approach to innovation management (Lawson and Samson 2001; O'Connor 2008), our insights offer important guidance about where to start, beyond the idea that everything is interrelated (Teece 2018). Our proposed approach emphasizes the importance of having a clear innovation strategy that enables adequate investments in innovation activities, including motivation for risk-taking (Giaccone and Magnusson 2022). Also, another key aspect is adopting a systemic view of innovation, which requires the use of a sufficiently comprehensive set of innovation management system elements, and recognition that these elements are aligned and support each other. Finally, even with an innovation management system in place, practitioners must recognize the difficulties inherent in measuring and controlling innovation and acknowledge that innovation is characterized by uncertainty (Reinertsen and Shaeffer 2005).

Conclusion

Designing and implementing an innovation management system is a promising path to accelerate innovation efforts and build innovation capabilities. There is no universal ready-to-implement model, but the ISO 56002 standard is a mechanism to strengthen an organization's innovation efforts. When organizations design their innovation management system, how the ISO 56002 standard is used can impact innovation outcomes. Organizations need to consider their specific innovation ambitions as well as the resources and staff competencies they can engage. Systemic dimensions regarding how the elements interact and align, along with striking a proper balance regarding how much structure to induce, are important considerations for developing an effective innovation management system that enhances innovation efforts. Practitioners can apply the key insights outlined to develop an innovation management system that will help them achieve their innovation objectives.

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ORCID

Ingrid Kihlander http://orcid.org/0000-0001-8790-3859
Mats Magnusson http://orcid.org/0000-0002-6024-7908
Magnus Karlsson http://orcid.org/0000-0002-6013-011X

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